# MICROSPHERE ASSISTED LIGHT-SCATTERING IMAGING OF PLASMONIC NANOPARTICLES AT THE SINGLE PARTICLE LEVEL

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## ABSTRACT

We present a microsphere assisted imaging approach for recording the scattered light of a single plasmonic nanoparticle. Upon the utilization of microspheres, single gold nanoparticle with diameter below the classic optical diffraction limit can be imaged under the conventional light microscope.

**KEYWORDS:** Plasmonic Nanoparticles, Single Particle, Scattered Light, Microsphere

## INTRODUCTION

Metallic nanoparticles support collective oscillation of the conduction band electrons induced by incident electromagnetic radiation, known as the phenomenon of localized surface plasmon resonance (LSPR). The acquisition and analysis of LSPR spectral information help to identify the physical parameters (i.e., the size, shape, material and the refractive index of its surroundings) of the plasmonic nanoparticles and have been widely used in many research directions, including surface enhanced Raman spectroscopy (SERS), biosensing, surface catalysis, plasmonic tags, particle tracing, among many others[1]. However, the scattering light that carries the LSPR spectral information of single plasmonic nanoparticle is relative weak and decreases dramatically with its size (dependence $\sim r^{-6}$). The conventional imaging approach such as dark-field microscope (DFM) requires long acquisition time, illumination sources of high power and objectives with high numerical aperture, which results in low time resolution and expensive instrument costs. Recent studies have shown that dielectric microspheres have the capability to generate high-intensity photonic hotspots and provide image resolution beyond diffraction limit[2]. Here in this study, we report for the first time that dielectric microspheres can be utilized to excite and collect the scattered light of the plasmonic nanoparticles and achieve the light-scattering imaging of single plasmonic nanoparticles under conventional light microscope. We describe the imaging setup and principle, and demonstrate the imaging of single gold nanoparticle of 100 nm.

## EXPERIMENTAL

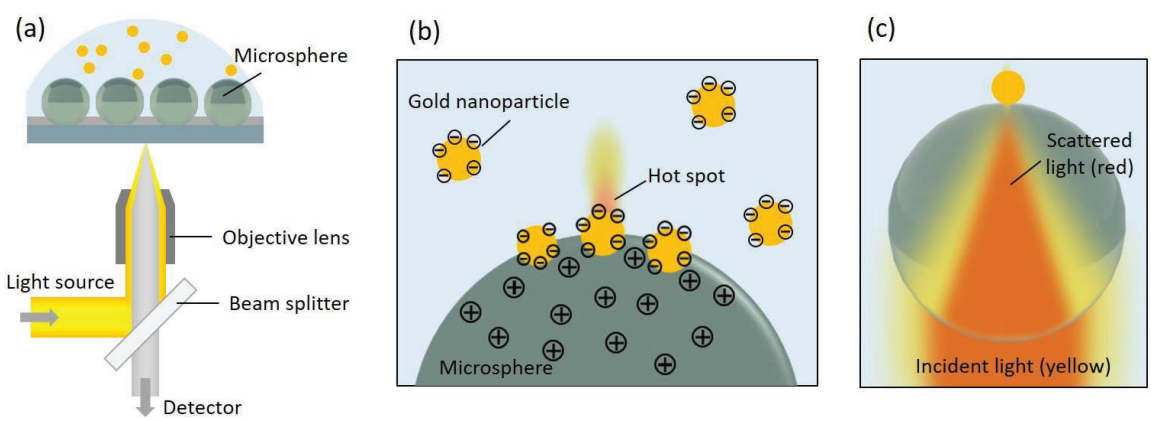


*Figure 1. (a) Schematic illustration of the optical configuration. Reflected light path is utilized. (b) The gold nanoparticles adhere on the microsphere due to electrostatic force. (c) Upon illumination, the photonic hot spot is generated and illumines on the gold nanoparticle. The back scattered light from gold nanoparticle is collected by the microsphere and finally recorded and imaged by the light microscope.*

Conventional barium titanate glass (BTG) dielectric microsphere (refractive index 1.9) with size of ~25 μm are coated on the glass substrate with a layer of polydimethylsiloxane (PDMS). The sample is modified with 3-aminopropyltriethoxysilane (APTES), resulting in positive surface charges. 50 microliter of citrate gold

nanoparticles with diameter of 100 nm are dropped on the dielectric microsphere lens. The sample is characterized under a conventional inverted microscope equipped with 40× objective (NA=0.55) and colored CCD camera under reflected light path, Figure 1. A LED white light source is utilized as the illumination source.

## RESULTS AND DISCUSSION

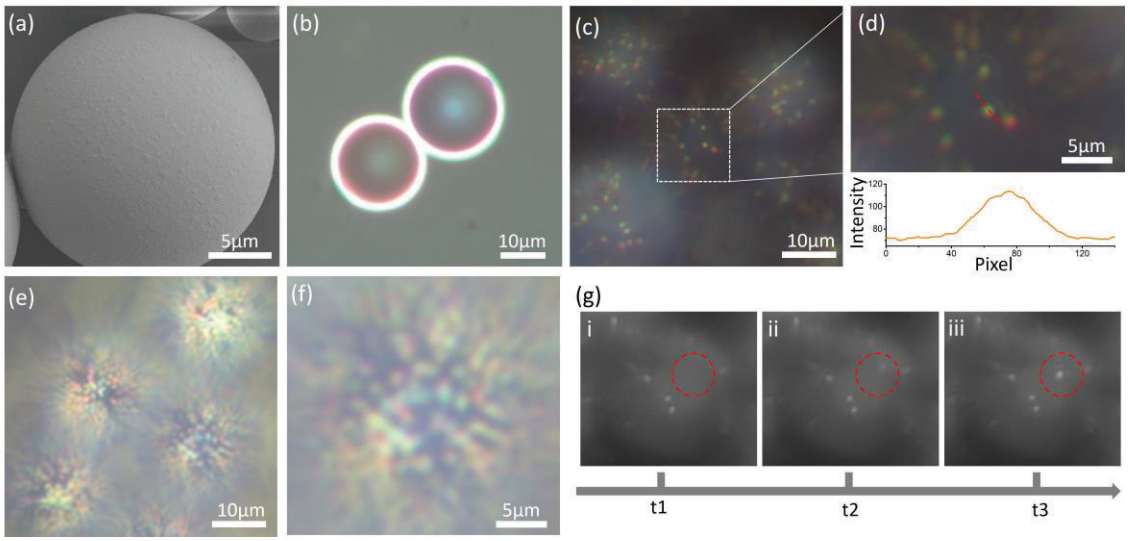


*Figure 2. (a) Scanning electron microscope (SEM) image of the BTG microsphere. (b) BTG microsphere in the solution under light microscope. (c) The light-scattering image of the gold nanoparticles. The bright spots indicate the nanoparticles. Image (c) shows the low-density nanoparticle adhesion on the BTG microsphere. (d) Enlarged view from (c) shows the single nanoparticle and its profile. (e) High density nanoparticle adhesion on the BTG microspheres and image (f) shows the enlarged view. (g) Real time monitoring of the nanoparticles adhesion process from a video. The red dashed circles in image (g)-i-iii indicate the area where a nanoparticle appears. At time t2, the nanoparticle appears and at time t3 the nanoparticle adheres on the microsphere.*

The BTG microspheres with refractive index ~1.9 generate photonic hotspots under the reflected light path. Their morphology is characterized under the SEM and light microscope, Figure 2(a) and (b). On the presence of the gold nanoparticle in the solution, adhered nanoparticles can be observed as shown in Figure 2(c)-(f). Single nanoparticle can be distinguished clearly at low-density adhesion, Figure 2(c) and (d). Specially, owing to the high time solution, the process of a single nanoparticle adhesion can be monitored at millisecond scale as shown in Figure 2(g).

## CONCLUSION

The light-scattering imaging method for plasmonic nanoparticles demonstrated here will assist in realizing of fast LSPR acquisition and high sensitive LSPR based biosensors, as well as providing a platform for investigating a wide range of plasmonic nanoparticle based fast interaction processes.

## ACKNOWLEDGEMENTS

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